GAS PROCESSING & METHANOL COMPLEX PROJECT NO. 11998A

Project	FEED for Gas Gathering Pipelines & Associated Infrastructure
Client	Brass Fertilizer and Petrochemical Company Limited
Originating Company	Epic Atlantic Limited
Document Title	**Pipe Stress Analysis Report-BBOU**
Document Number	EA-BFPCL-GPMC-GGPL-FEED-MPP-RPT-002
Document Revision	R01
Document Status	Issued for Review
Originator	Israel Osobase
Security Classification	Restricted
Issue Date	**09-Aug-22**

Pipe Stress Analysis Report-BBOU

R01	09.08.22	Issued for Review	I. Osobase	E. Nwachukwu	A. Duru	
Rev #	Issue Date	Status Description	Originator	Reviewer	Approver (Epic)	Approver (BFPCL)

Revision History

Rev No.	Date	Description of Revision
R01	09.08.22	Issued for Review

Revision Philosophy:

a. All documents for review shall be issued at R01, with subsequent R02, R03, etc as required.

b. All documents approved for issue or approved for design shall be issued at A01 with subsequent A02, A03, etc. as required. (Management of Change is required for A02, A03, etc.)

for the subsequent project phases, post FEED, it is expected that:

c. All Detailed Design documents for review shall be issued at D01, with subsequent D02, D03, etc. as required.

d. All documents approved for construction shall be issued at C01 with subsequent C02, C03, etc. as required. (Management of Change is required for C02, C03, etc.)

e. All approved "As-Built" documents shall be issued at Z01, with subsequent Z02, Z03, etc as required. (Use versions Z01.1, Z01.2, Z01.3, etc to review the "As-Built" document to Z02).

SUMMARY PAGE

Client:	Brass Fertilizer & Petrochemical Company Limited	
Project:	Gas Processing & Methanol Complex	**Project No.** 11998A
Facility:	Gas Gathering Pipelines & Associated Infrastructure	**Doc No.** EA-BFPCL-GPMC-GGPL-FEED-MPP-RPT-002

The administration of this document is in respect of FEED for Gas Gathering Pipelines & Associated Infrastructure for the Gas Processing & Methanol Plant Project of BFPCL.

The Pipelines will be implemented in two phases, - starting with BBOU & ELEP as Phase 1and followed by NUNR, SEIB & OPUG as Phase 2. The focus of the present FEED is on Phase 1, while Phase 2 will be implemented later.

It is also a precursor and accompaniment to the FEED study & report that will lead to the project's FID and implementation.

The report has relied on and used data and reference documents provided by the client, and if further documents and information become available, Epic Atlantic reserves the right to update the opinion set out in this report.

For better appreciation, this document should be read in conjunction with the references listed below, which are the Concept Select Study reports that constitute basis for the Conceptual Study.

KEYWORDS: BASIS OF DESIGN, PIPELINE, MANIFOLD, PLATFORM, MULTIPHASE FLOW, ROW, PTS

Table of Contents

1. Introduction

1.1. Background Information

Brass Fertilizer and Petrochemical Company Limited (BFPCL) is developing a 10,000 metric tonnes per day (MTPD) capacity greenfield methanol production facility in Akabel (Odioma), Brass LGA, Bayelsa state. The plant complex is to be based on natural gas feedstock from 5 SPDC fields in OMLs 33, 35 & 36.

The Project involves the development, construction, and operation of a plant comprising:

- Two trains of 5,000 MTPD methanol plant, producing a total of 10,000 MTPD of methanol
- 340 MMscfd Gas Processing Plant ("GPP"), which will extract NGL from the natural gas feedstock, prior to feeding the lean gas to the methanol plant.
- Gas gathering pipelines and associated manifolds transporting the natural gas from five fields, viz: Bubouwe Bou (BBOU), Elepa (ELEP), Nun River (NUNR), Seibou (SEIB), and Opugbene (OPUG) to the GPP at Akabel.
- Product Export Lines and SBM for the export of the methanol, and NGL of the GPP.
- All offsite, utility, and off-plot facilities (including product storage, internal power generation, steam generation, and export jetty) for the operation of the processing plants.

The Pipelines will be implemented in two phases, - starting with BBOU & ELEP as Phase 1 and followed by NUNR, SEIB & OPUG as Phase 2.

The focus of the present FEED is on Phase 1 as shown in figure 1 below, while Phase 2 will be implemented later.

TCE is the Project Management Consultant for the project.

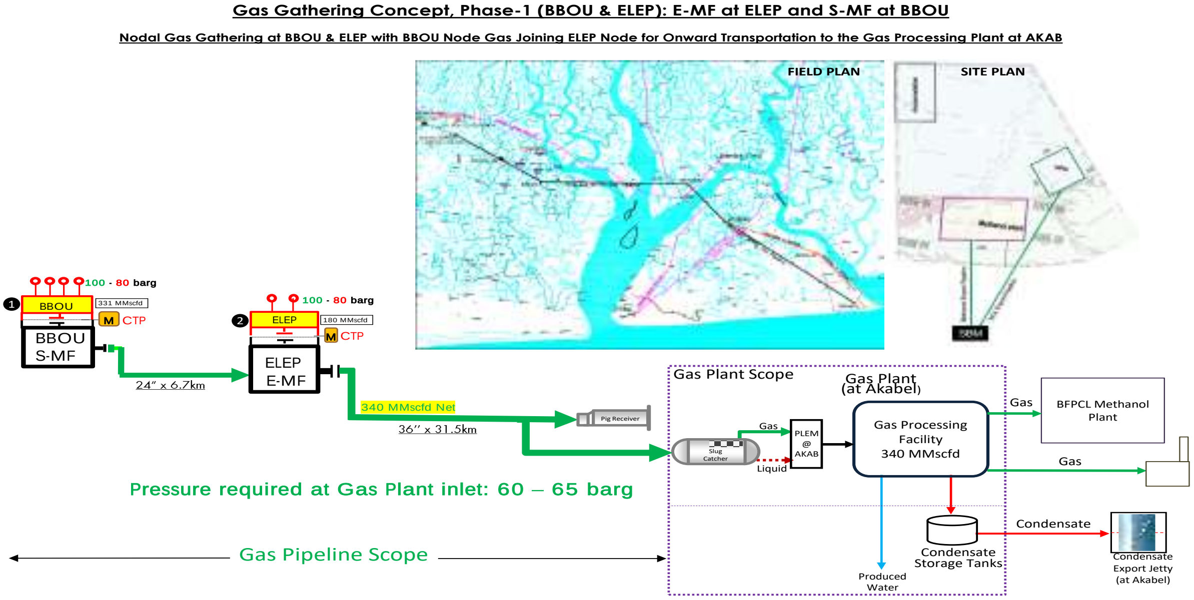


Figure 1: Gas Gathering Concept, Phase 1

1.2. Purpose

The purpose of this document is to present the result of the stress analysis carried out on the pipeline and piping systems of Bubouwe Bou Facility.

The analysis is based on the methodology for the computerized stress analysis.

The governing code for this analysis is piping code ASME B31.3 and B31.8.

The following shall be ensured;

- That the stresses induced are within the allowable limits as per the applicable code.
- To ensure that it has inherent flexibility to accommodate thermal displacements due to design/operating conditions without imposing excessive loads on pipe supports and equipment nozzles limits as per the applicable code.
- That the stresses induced are within the allowable limits as per the applicable code.
- The piping system behaviour is within permissible limits for the given design parameters in the given piping layout and terminal boundary conditions.
- The piping system is adequately supported to prevent vibration, excessive displacement and system resonance

1.3. Definition of Terms

ABBREVIATIONS	MEANING
AED	Anti-Explosive Decompression
ASME	American Society of Mechanical Engineers
ASTM	American Society for Testing and Materials
API	American Petroleum Institute
BS	British Standard
CS	Carbon Steel
DBB	Double Block and Bleed
ENP	Electroless Nickel Plated
HFE	Human Factor Engineering
ISO	International Organisation for Standardization.
MESC	Material & Equipment Standard and Code
MMscf/d	Million standard cubic feet per day
MSS	Manufacturers Standardisation Society
MTO	Material Take Off
NDE	Non-Destructive Examination

NPT	Nominal Pipe Threaded
NPS	Nominal Pipe Size
RTJ	Ring Type Joint
SPE	Society of Petroleum Engineers.
SS	Stainless Steel
SW	Socket Weld

2. Regulations, Codes And Standards

The latest versions of the regulations, codes, and standards and extant amendment or supplements shall apply. Local laws regulating the oil and gas industry in Nigeria shall be strictly adhered to. The order of precedence for the documents shall be as follows:

- Nigerian Local Standards and Codes
- The Contract
- Project Specifications
- Employer's Standards
- Shell DEPs
- International Codes and Standards

In all cases, the most stringent shall apply except where constraints dictate otherwise.

Conflicts between specifications and referenced Codes and Standards or deviations thereof shall be referred in a Technical Query (TQ) to the EMPLOYER for resolution.

2.1. Project Technical Documents

Key Project Technical Specifications are listed below.

REF	DOCUMENT NUMBER	TITLE
[1]	EA-BFPCL-GPMC-GGPL-FEED-GEN-RPT-002	Basis of Design
[2]	EA-BFPCL-GPMC-GGPL-FEED-MPP-SPC-005	Piping Class Specification
[3]	EA-BFPCL-GPMC-GGPL-FEED-MPP-SPC-015	Specification for Piping
[4]	EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003	Process Engineering Flow Sheets (PEFS) Elep Facilty
[5]	EA-BFPCL-GPMC-GGPL-FEED-MPP-DWG-015	General Piping Arrangement Drawing - ELEPA
[6]	EA-BFPCL-GPMC-GGPL-FEED-MPP-DWG-024	Pipe Supports Location Drawing - ELEPA
[7]	EA-BFPCL-GPMC-GGPL-FEED-MPP-LST-012	Critical Line List- ELEPA

2.2. Shell Specifications

The following DEPs are applicable:

	DOCUMENT NUMBER	TITLE
[8]	DEP 31.38.01.25-Gen	Piping -Process Design Requirements
[9]	DEP 31.38.01.10-Gen	Piping classes – basis of design
[10]	DEP 31.38.01.14-Gen	Piping classes
[11]	DEP 31.38.01.24-Gen	Piping – Engineering and Layout Requirements
[12]	DEP 31.38.01.26-Gen	Piping – Pipe Stress Analysis Requirements

2.3. International Codes and Standards

REF	DOCUMENT NUMBER	TITLE
[13]	ASME B31.8	Gas Transmission and Distribution Piping Systems
[14]	ASME B31.3	Process Piping Code
[15]	ASME B36.10M	Welded and Seamless Wrought Steel Pipe
[16]	API 5L	Specification for Line Pipe
[17]	ASME B16.5	Steel Pipe Flanges NPS ½ through NPS 24
[18]	ASME B16.20	Metallic Gaskets for Pipe Flanges
[19]	ASME B31.3	Process Piping Code
[20]	ASME B16.5	Steel Pipe Flanges NPS ½ through NPS 24
[21]	ASME B16.20	Metallic Gaskets for Pipe Flanges

2.4. System of Units

In general, the SI system of units shall be used for the Project, with the following exceptions:

- Pipe spanning shall be in millimeters
- Flange ratings shall be in pounds as per ASME B16.5
- Pressure shall be in bar.
- Pipe size shall be in inches.
- All other units are based on metric standard unit system.

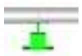
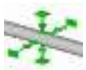
2.5. Safety

Safety is the primary consideration during design and all risks associated with the design shall be minimized by adhering to the relevant governing legislation, regulations, design codes, and accidental procedural requirements.

3. Definition of Concepts

3.1. Support Definitions

Definition of restraints used are given bellow. The effect of frictional resistance to steel surfaces in contact is considered by a static friction coefficient of 0.3 (steel / steel)

symbol	Restrain and support type
	V-stop or Rest supports restrain the pipe in vertical direction. Acts only down-wards
	Guide supports restrain all directions expect in axial direction of the pipe. Modelled with a gap of 3mm in all directions except downwards to allow some unrestricted thermal expansion.
	Anchor Restrains pipe movement in all direction.

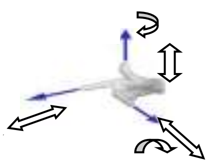
3.2. Geometry and system Model

The geometry used for the modelling of the piping system analyzed is based on latest PDMS 3D model

A unique node number was assigned to all points in the model indicating where there is a support, piping items like valve, transition point, change in diameter, wall thickness, material, temperature, or pressure, or a change in direction.

3.3. Coordinate System and Degrees of Freedom

The coordinate system for this AutoPIPE stress analysis model is the right-hand rule, with the +Y direction pointing upwards, X direction parallel to the pipeline axis and Z direction perpendicular to the afore - mentioned axis. See Figure below for coordinate system. Each location has six degrees of freedom, represented with three translational movements X, Y and Z and three rotational movements RX, RY and RZ.



4. Analysis Basis and Assumptions

4.1. General

- The ratio of calculated stress with material allowable stress shall be less than 1.0 in all cases.
- The basis for stress analysis, loading methods are specified in the technical specification for piping stress analysis and approved by client prior to the commencement of this analysis.
- Software for the computerized stress analysis shall be Bentley AutoPIPE Advanced version 12.5 or higher versions
- The loads considered in this analysis are pressure, temperature, weight (live and dead loads), wind, vibration reactions, thermal expansion and contraction, temperature gradients. However, earthquake-induced horizontal and vertical forces are not considered.
- There are no dynamic conditions for the piping system. Surge, slug, seismic and other forms of vibrations are not considered.
- The occasional effect of wind is considered with a projected application method, and a ground datum of 1m. No wind load shall be applied below this elevation. The wind speed is 35.6m/s
- For hydrotest load case, a pressure factor of 1.5 shall be applied on the design pressure in accordance with ASME B31.3 for Process Piping while a pressure factor of 1.25 shall be applied on design pressure as per ASME B31.8 for Pipelines and Pig Receiver
- This report presents analysis for lines classified as piping (per ASME B31.3) and Pipeline (per ASME B31.8) in code break of P&IDs (EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003). For the purpose of this analysis, the system is anchored at all interfaces between piping and pipeline to achieve individual system reaction to service loads as per governing codes.
- Pipe routing is based on the Latest PDMS 3D Model routing.
- Flange and valve Loads (weight) are taken from the software library catalogue.
- All piping materials as identified in Specification for Piping Material Class
- The minimum natural frequency of the system shall be maintained at a threshold limit of 3 Hz.

4.2. Boundary Condition

All lines within Bubouwe Bou facility shall be considered not buried in this analysis, therefore active length required to form a virtual anchor has been calculated and it shall span from interface between buried and above ground pipe to the point of anchorage for both ends to get realistic boundary conditions of the system.

4.3. Virtual Anchor

Virtual anchor has been considered for the buried segments of the 24″ pipeline, the system was modelled with a calculated virtual anchor generated at node A00/BA02 with length of 373m .. The detailed calculation and input data are in Attachment A

5. Input Data

In addition to the following data, input data is automatically generated by Autopipe program and form part of the analysis report. See Attachment A

5.1. Lines Analysed

PIPELINE (ASME B31.8 SCOPE)

- BBOU-24-PL-012-X70-CL900_UG
- BBOU-A-100
- BBOU-A-200
- BBOU-24-PL-013-X70-CL900_AG
- BBOU-24-PL-013-X70-CL900_AG_B
- BBOU-24-PL-012-X70-CL900_AG
- BBOU-24-PL-013-X70-CL900_UG

PIPING (ASME B31.3 SCOPE)

- BBOU-8-P-014-C03
- BBOU-24-P-013-C03
- BBOU-4"-P-016-C06
- BBOU-24-P-008-C03
- BBOU-24-P-008-C03_BYPASS
- BBOU-24-P-003-C03
- BBOU-24-PF-XXXX-120-E
- BBOU-6-B-005-B06
- BBOU-4"-P-016-C06_B
- BBOU-6-B-005-B06_B
- BBOU-A-300-N7
- BBOU-4"-P-021-C06
- BBOU-XX-B-007-B06
- BBOU-4"-P-021-C06_B

- BBOU-XX-B-007-B06_B
- BBOU-6-P-006-C06
- BBOU-6-B-003-B06
- BBOU-6-P-007-C06
- BBOU-6-B-004-B06
- BBOU-6-B-001-B06
- BBOU-8-P-008-C03

5.2. Environmental data

Parameters	Values
Ambient Temperature	23°C
Prevailing wind direction	south-west
Basic wind speed	35.6 m/s
Ground elevation for wind load application	From grade
Wind shape factor (referred as force coefficient per ASCE)	1
Wind specification type	ASCE-7-10*
Wind exposure	Category C as per
Wind application method	Projected **
Wind direction	Global +X, -X, +Z , -Z

*ASME B31.3 section 301.5.2 refers wind load analysis of exposed piping to the procedures in ASCE 7

**The projected area of pipe perpendicular to the direction of the wind is used to determine the magnitude of the wind force. The wind force is then applied to the pipe in the direction of the wind.

5.3. Friction Coefficient

The coefficient of friction between supporting interface (i.e., the junction between Top of Steel and Bottom of Pipe) shall be applied at all vertical restraints (Ref. Specification for stress analysis)

		Friction Coefficient

Material in Con-	Surface Condi-	Static	(Sliding)
Metal on metal	Dry	0.5-0.8	0.4
Teflon on metal	Dry	0.05-0.2	0.1

5.4. Load definitions

S/No	Load case	Description
1	Gr	Dead weight of pipe and content
2	T1	Max design temperature, 60/100 ºC
3	T2	Operating temperature, -29/12ºC
4	T3	Ambient temp. 23 ºC
5	P1	Design pressure, 1.96 /11 N/mm2,
6	P2	Max Operating pressure, 10 N/mm2
7	Hydro.P	Hydrotest pressure; 1.5 * P1 @T3, 1.25*P1 @ T3
8	Wind 1	Projected Wind load in Global +X direction @ basic wind speed at 38 m/s, from 2m datum analyzed to ASCE-7-2010
9	Wind 2	Projected Wind load in Global -X direction @ basic wind speed at 38 m/s, from 2m datum analyzed to ASCE-7-2010
10	Wind 3	Projected Wind load in Global +Z direction @ basic wind speed at 38 m/s, from 2m datum analyzed to ASCE-7-2010
11	Wind 4	Projected Wind load in Global -Z direction @ basic wind speed at 38 m/s, from 2m datum analyzed to ASCE-7-2010
12	G	Gravity

5.5. Load Cases Considered

The following single and combination load sequence are considered for AutoPipe non-linear analysis of stress, moments and forces result at all nodal point.

5.6. Load Combinations

Load Combination

New | Modify | Duplicate Current | Delete Selected | Delete All | Reset Defaults Only

Combination	Category	Print	Auto Update	Combination Method	Combination Type	Allowable (K) Factor	Allowable Stress N/mm2	Long Allow N/mm2	Total Allow N/mm2	Description
		✓	✓							
GR + Max P{1}	Sustain	✓	✓	Sum	Default		Automatic			
Max Range	Expansion	✓	✓	Sum	Default		Automatic			
Amb to T1{1}	Expansion	✓	✓	Sum	Default		Automatic			
Amb to T2{1}	Expansion	✓	✓	Sum	Default		Automatic			
T1 to T2{1}	Expansion	✓	✓	Sum	Default		Automatic			
Sus. + W1{1}	Occasion	✓	✓	Abs. Sum	Default	0.75	Automatic			
Sus. + W2{1}	Occasion	✓	✓	Abs. Sum	Default	0.75	Automatic			
Sus. + W3{1}	Occasion	✓	✓	Abs. Sum	Default	0.75	Automatic			
Sus. + W4{1}	Occasion	✓	✓	Abs. Sum	Default	0.75	Automatic			
Max P{1}	Hoop	✓	✓	Sum	Default		Automatic			
GRTP1{1}	Rest-Func	✓	✓	Sum	Default			Auto	Auto	
GRTP1+W1{1}	Rest-Env	✓	✓	Sum	Default			Auto	Auto	
GRTP1-W1{1}	Rest-Env	✓	✓	Sum	Default			Auto	Auto	
GRTP1+W2{1}	Rest-Env	✓	✓	Sum	Default			Auto	Auto	
GRTP1-W2{1}	Rest-Env	✓	✓	Sum	Default			Auto	Auto	
GRTP1+W3{1}	Rest-Env	✓	✓	Sum	Default			Auto	Auto	
GRTP1-W3{1}	Rest-Env	✓	✓	Sum	Default			Auto	Auto	
GRTP1+W4{1}	Rest-Env	✓	✓	Sum	Default			Auto	Auto	
GRTP1-W4{1}	Rest-Env	✓	✓	Sum	Default			Auto	Auto	
GRTP2{1}	Rest-Func	✓	✓	Sum	Default			Auto	Auto	
GRTP2+W1{1}	Rest-Env	✓	✓	Sum	Default			Auto	Auto	
GRTP2-W1{1}	Rest-Env	✓	✓	Sum	Default			Auto	Auto	
GRTP2+W2{1}	Rest-Env	✓	✓	Sum	Default			Auto	Auto	
GRTP2-W2{1}	Rest-Env	✓	✓	Sum	Default			Auto	Auto	
GRTP2+W3{1}	Rest-Env	✓	✓	Sum	Default			Auto	Auto	
GRTP2-W3{1}	Rest-Env	✓	✓	Sum	Default			Auto	Auto	
GRTP2+W4{1}	Rest-Env	✓	✓	Sum	Default			Auto	Auto	
GRTP2-W4{1}	Rest-Env	✓	✓	Sum	Default			Auto	Auto	
Hydrotest-NL{2}	Occasion	✓	✓	Sum	Default	0.90	Automatic			

Load Combination

New | Modify | Duplicate Current | Delete Selected | Delete All | Reset Defaults Only

Combination	Print	Auto Update	Combination Method	Combination Type	Description
	✓	✓			
Gravity{1}	✓	✓	Sum	Default	
Thermal 1{1}	✓	✓	Sum	Default	60.00 deg C
Thermal 2{1}	✓	✓	Sum	Default	38.10 deg C
Wind 1{1}	✓	✓	Sum	Default	0.00, 0.00, -1.00
Wind 2{1}	✓	✓	Sum	Default	0.00, 0.00, 1.00
Wind 3{1}	✓	✓	Sum	Default	1.00, 0.00, 0.00
Wind 4{1}	✓	✓	Sum	Default	-1.00, 0.00, 0.00
Pressure 1{1}	✓	✓	Sum	Default	
Pressure 2{1}	✓	✓	Sum	Default	
GT1{1}	✓	✓	Sum	Default	
GT2{1}	✓	✓	Sum	Default	
GW1{1}	✓	✓	Sum	Default	
GW2{1}	✓	✓	Sum	Default	
GW3{1}	✓	✓	Sum	Default	
GW4{1}	✓	✓	Sum	Default	
GT1P1{1}	✓	✓	Sum	Default	
GT2P2{1}	✓	✓	Sum	Default	

See attachment A for a full list of AutoPIPE load cases

6. Result Summary

In the following tables, calculated stress values at Nodal points are reported.

For each tabulation, the maximum stress is given, the point at which it occurs, the load combination, and the ratio between calculated and code allowable stresses.

6.1. Stress Summary

Analysis for pipeline and piping section was included in this report, and the calculation was done in accordance with ASME B31.8 and ASME B31.3, respectively.

The ASME B31.8 scope refers to pipeline section comprising underground and aboveground pipelines including the 24″ Pig Receiver and Launcher (BBOU-A-100& BBOU-A-200).

The ASME B31.3 scope refers to piping systems, Manifold header and vent lines.

For each section, the basis of analysis and findings will be as per the applicable code.

Such that in the case of applicability of ASME B31.3, only results in piping section will be taken into consideration and that of the pipeline section will be ignored. Same applies for ASME B31.8 on the pipeline section.

6.2. ASME B31.3 Maximum code Stresses

Stress Category	Nodal Point	Max. Stress Value (N/mm2)	Allowable Stress (N/mm2)	Ratio	Load Combination
Maximum sustained stress	AR06	77	138	0.56	GR + Max P
Maximum displacement stress	T10	197	207	0.95	Max Range
Maximum occasional stress	Z01	108	403	0.27	Hydrotest-NL
Maximum hoop stress	Y00	154	206	0.75	Max P
Maximum sustained stress ratio	AR06	77	138	0.56	GR + Max P
Maximum displacement stress ratio	T10	197	207	0.95	Max Range
Maximum occasional stress ratio	E08	103	217	0.48	Hydrotest-NL
Maximum hoop stress ratio	Y00	154	206	0.75	Max P

The system is within the allowable stresses check for all load cases.

6.3. ASME B31.8 Maximum code Stresses

Stress Category	Nodal Point	Max. Stress Value (N/mm2)	Allowable Stress (N/mm2)	Ratio	Load Combination
Maximum sustained stress	P12	176	362	0.49	GR + Max P
Maximum displacement stress	AZ02N	269	303	0.89	Max Range
Maximum occasional stress	P12	260	434	0.60	Hydrotest-NL
Maximum hoop stress	A00	211	347	0.61	Max P
Maximum sustained stress ratio	P12	176	362	0.49	GR + Max P
Maximum displacement stress ratio	P12	263	290	0.90	Max Range
Maximum occasional stress ratio	P12	260	434	0.60	Sus. + W4
Maximum hoop stress ratio	A00	211	347	0.61	Max P

The system is within the allowable stresses check for all load cases.

6.4. Maximum Displacements

Direction	Displacement(mm)	Node	Load combination
Maximum X	29	AZ00	GT1P1
Maximum Y	69	P14	GT1P1
Maximum Z	-1	W04	GT1P1

6.5. Maximum Restraint Forces

Direction	Restraint forces (N)	Node	Load combination
Maximum X	-365349	A00	GT1P1
Maximum Y	-449209	AZ04	GT2
Maximum Z	-18336	W02	P2

7. Conclusion

In conclusion, the following has been established for the system:

a) Results show all calculated stresses (sustained, expansion and occasional cases) are within allowable limits as per ASME B31.8

b) Results show all calculated stresses (sustained, expansion and occasional cases) are within allowable limits as per ASME B31.3.

c) The first natural frequency of both systems is above 3Hz as specified in section 4.1

8. List of Attachments

Attachment A; Input Listing

(Component data, material properties and allowable stress values, Load cases description, Hydrotest data, Temp and pressure, Soil parameters, other input data.)

ASME_B31.3_INPUT LISTING.pdf

ASME_B31.8_INPUT LISTING.pdf

Attachment B; Analysis results

ASME_B31.3_ANALYSIS RESULT.pdf

ASME_B31.8_ANALYSIS RESULT.pdf

Attachment C; Screen shot from Autopipe

SCREEN_Shot.BBOU.pdf

Attachment D; Stress Isometrics

ASME_B31.3_STRESS_ISO.pdf

ASME_B31.8_STRESS_ISO.pdf